



Connect a GitHub Repo with AWS



Muhammad Sadique

```
TERMINAL  PROBLEMS  OUTPUT  DEBUG CONSOLE  PORTS
[ec2-user@ip-172-31-28-234 nextwork-web-project]$ sudo dnf update -y
sudo dnf install git -y
Last metadata expiration check: 0:43:16 ago on Thu Oct 10 22:01:15 2024.
Dependencies resolved.
Nothing to do.
Complete!
Last metadata expiration check: 0:43:17 ago on Thu Oct 10 22:01:15 2024.
Dependencies resolved.
=====
Package                                Architecture  Version                                Repository  Size
Installing:
git                                     x86_64        2.40.1-1.amzn2023.0.3                 amazonlinux  54 k
Installing dependencies:
git-core                                x86_64        2.40.1-1.amzn2023.0.3                 amazonlinux  4.3 M
git-core-doc                            noarch        2.40.1-1.amzn2023.0.3                 amazonlinux  2.6 M
perl-Error                              noarch        1:0.17029-5.amzn2023.0.2             amazonlinux  41 k
perl-File-Find                          noarch        1.37-477.amzn2023.0.6                 amazonlinux  26 k
perl-Git                                noarch        2.40.1-1.amzn2023.0.3                 amazonlinux  42 k
=====
SSH: ec2-34-211-81-163.us-west-2.compute...
```



Introducing Today's Project!

Today we will setup a GitHub repository for our code to be stored in the cloud.

Key tools and concepts

Services i used were Git (Local) GitHub Key concepts that i learned were How to setup local git repo How to setup repo on GitHub How to push your code from local git repo to GitHub

Project reflection

The project took me approximately 120 minuts. The challanging part was to Generate the access token and using it.

I did this project to learn about CI/CD Pipeline. I surely meet my goal which is DevOps.

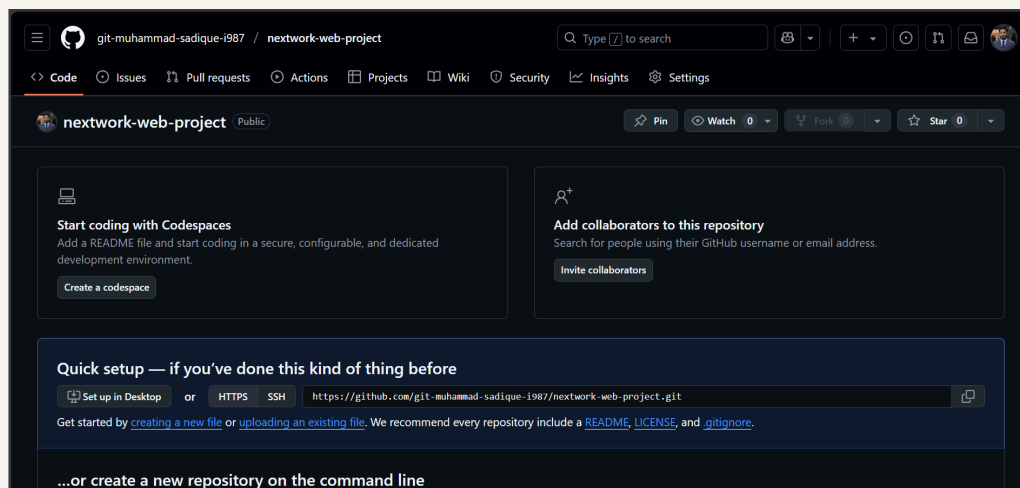
This project is part two of a series of DevOps projects where I'm building a CI/CD pipeline! I'll be working on the next project tomorrow - InshaALLAH.



Git and GitHub

Git is a version control system since it tracks your changes by taking snapshots of what your files look like at specific moments, and each snapshot is considered a 'version'. I installed it by running following commands on my EC2 instance terminal `sudo dnf update -y sudo dnf install git -y` to verify, I ran the following command `git --version`

GitHub is a shared code repository for developers to work in teams. I am using GitHub to share my updates and reviews to a shared code base.



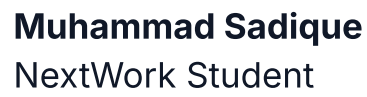


My local repository

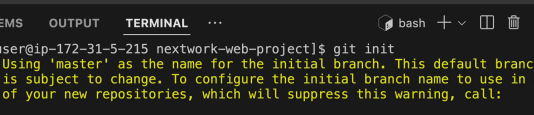
To store your code using Git, you create repositories (aka 'repos'), which are folders that contain all your project files and their entire version history. Hosting a repo in the cloud, like on GitHub, means you can also collaborate with other engineers and access your work from anywhere.

git init creates a local git repo in your ec2 instance which is not yet sync with your GitHub repo. Any change here will not reflect into your GitHub repo unless you explicitly commit the changes into your GitHub repo.

A branch is like a paralall universe in your Git repo to test code changes without messing with your main branch.



nextwork.org



The screenshot shows a terminal window with a dark background. At the top, there's a title bar with a search icon, a refresh icon, and a settings icon. Below the title bar, there's a navigation bar with tabs for 'PROBLEMS', 'OUTPUT', 'TERMINAL', and '...'. The 'TERMINAL' tab is active. The terminal content shows a user at a prompt on a remote host, initializing a Git repository. The user enters 'git init', followed by 'git config --global init.defaultBranch <name>', and then 'git branch -m <name>'. The terminal output provides instructions on branch naming and shows the repository being initialized. At the bottom of the terminal, there's a status bar with the text 'SSH: ec2-3-131-38-227.us-east-2.com' and a 'Cursor Tab' label.

```

[ec2-user@ip-172-31-5-215 nextwork-web-project]$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:     git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:     git branch -m <name>
Initialized empty Git repository in /home/ec2-user/nextwork-web-project/.git/
[ec2-user@ip-172-31-5-215 nextwork-web-project]$
  
```



To push local changes to GitHub, I ran three commands

git add

The first command i ran was `git add .` stages all (marked by the '.') files in `nextwork-web-project` to be saved in the next version of your project

git commit

`git commit -m "Updated index.jsp with new content"` saves the staged changes as a snapshot in your project's history. This means your project's version control history has just saved your latest changes in a new version. `-m` flag lets you leave a message describing what the commit is about, making it easier to review what changed in this version.

git push

Finally i ran the command `git push -u origin master` uploads i.e. 'pushes' your committed changes to origin, which you've bookmarked as your GitHub repo. 'master' tells Git that these updates should be pushed to the master branch of your GitHub repo. By using `-u` you're also setting an 'upstream' for your local branch, which means you're telling Git to remember to push to master by default. Next time, you can simply run `git push` without needing to define origin and master.



Authentication

When I commit changes to GitHub, Git asks for my credentials because, Git needs to double check that you have the right to push any changes to the remote origin your local repo is connected with. To do this, Git is now authenticating your identity by asking for your GitHub credentials.

Local Git identity

Git needs author information for commits to track who made what change. If you don't set it manually, Git uses the system's default username, which might not accurately represent your identity in your project's version history.

Run git log to see your history of commits, which also mentions the commit author's name.

```
[ec2-user@ip-172-31-28-234 nextwork-web-project]$ git log
commit 5b835879d52696057f37479109cdef879b82039a (HEAD -> master)
Author: EC2 Default User <natasha@nextwork.org>
Date: Sat Oct 12 05:26:39 2024 +0000
```

```
Updated index.jsp with new content
```



GitHub tokens

GitHub authentication failed when I entered my password because, GitHub phased out password authentication to connect with repositories over HTTPS - there are too many security risks and passwords can get intercepted over the internet ☹️ You need to use a personal access token instead, which is a more secure method for logging in and interacting with your repos.

A GitHub token is proof of access right granted for GitHub repo. Code push from local git repos can only be sync with GitHub repo if the user has the access token.

I could set up a GitHub token by going to settings > Developer Settings > Personal Access Token > Token (Classic) I generated the token and used it as password when i asked to.



Select scopes

Scopes define the access for personal tokens. [Read more about OAuth scopes.](#)

☒ repo	Full control of private repositories
☐ repo:status	Access commit status
☐ repo_deployment	Access deployment status
☐ public_repo	Access public repositories
☐ repo:invite	Access repository invitations
☐ security_events	Read and write security events
☐ workflow	Update GitHub Action workflows
☐ write:packages	Upload packages to GitHub Package Registry
☐ read:packages	Download packages from GitHub Package Registry
☐ delete:packages	Delete packages from GitHub Package Registry
☐ admin:org	Full control of org and teams, read and write org projects



Making changes again

I wanted to see Git working in action, so I updated index.jsp file by adding an additional line to it. I couldn't see the changes in my GitHub repo initially because after executing commands like: `git add .` `git commit -m "Add new line to index.jsp"` `git push` I had to provide my username and access token granted by GitHub to the ec2 terminal.

I finally saw the changes in my GitHub repo after entering my GitHub username and access token.

EC2 Default User Add new line to index.jsp

Code Blame 13 lines (7 loc) · 227 Bytes Code 55% faster with GitHub Copilot

```
1 <html>
2
3 <body>
4
5 <h2>Hello <YOUR NAME>!</h2>
6
7 <p>This is my NextWork web application working!</p>
8
9 <p>If you see this line in Github, that means your latest changes are getting pushed to your cloud repo :o</p>
10
11 </body>
12
13 </html>
```



nextwork.org

The place to learn & showcase your skills

Check out nextwork.org for more projects

